

ECEN 607: Advanced Analog Circuit Tech

Assignment:4

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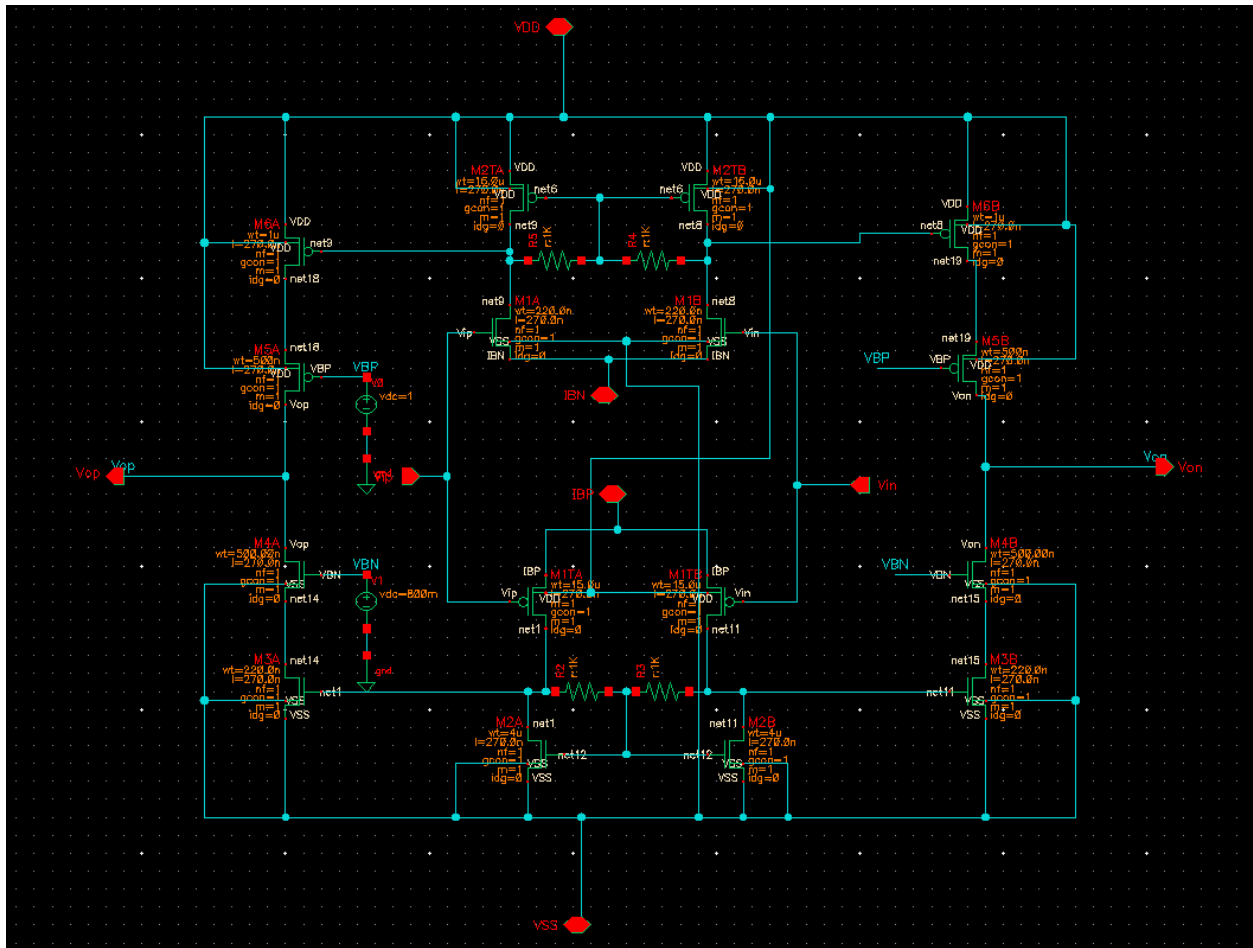
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Repeat your Assignment 3, but use the following architecture:
This design is fully-differential.

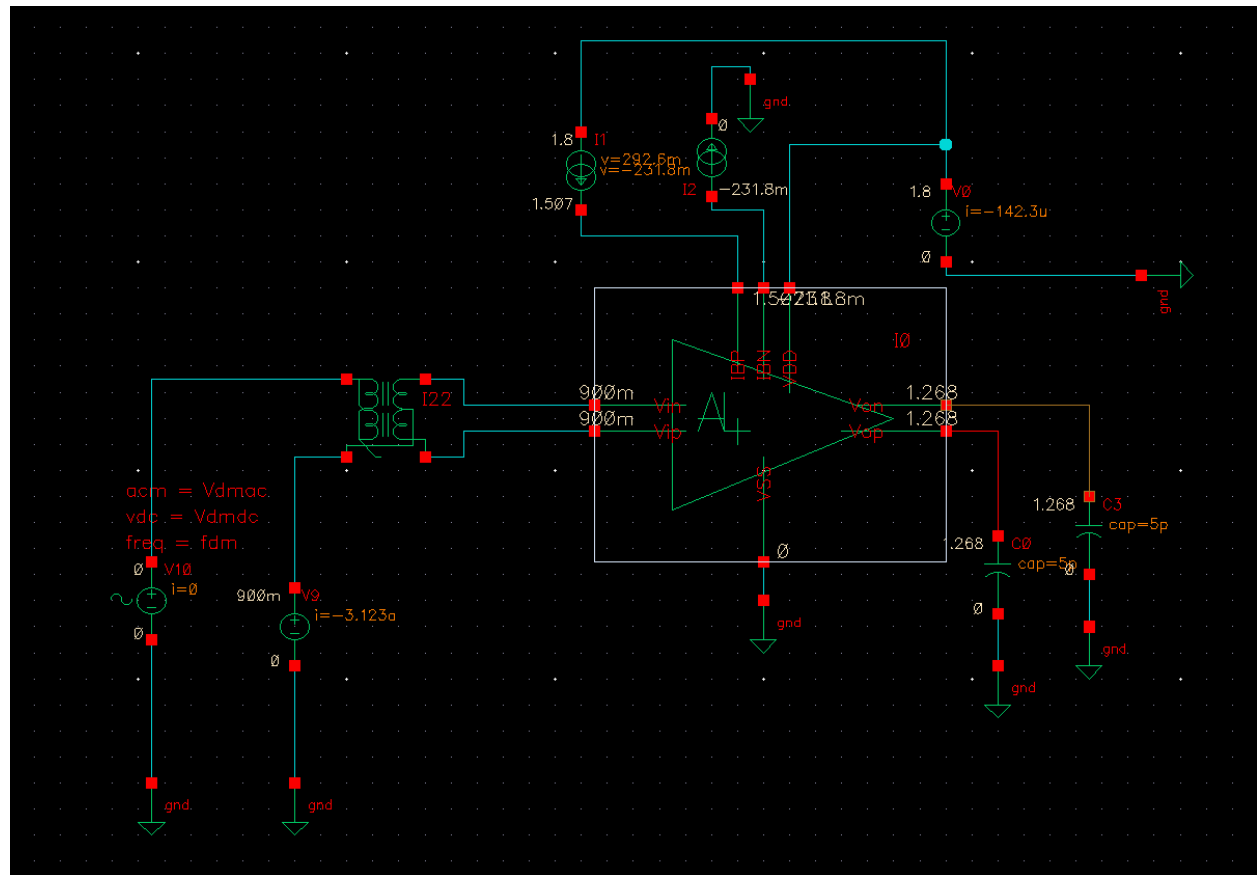
Reference: <https://ieeexplore.ieee.org/abstract/document/11165015/metrics#metrics>

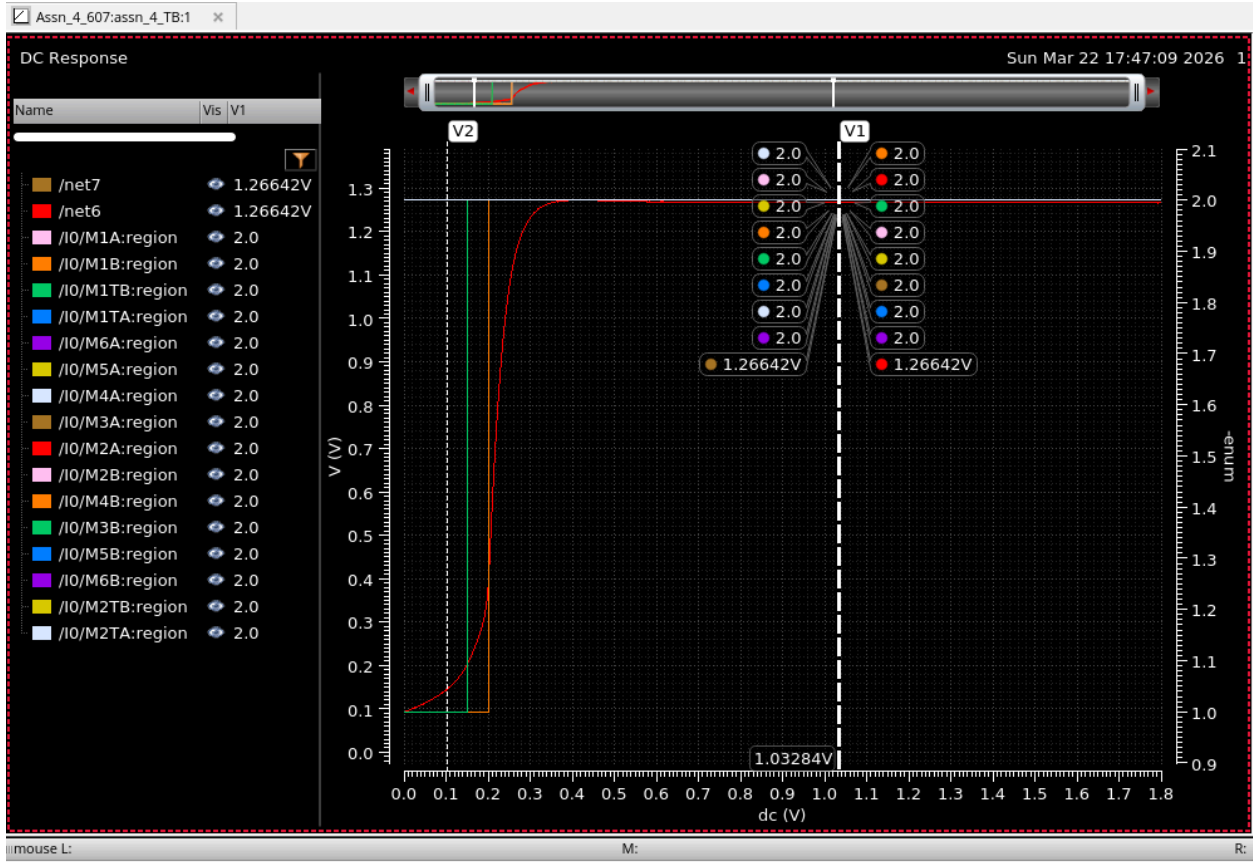
Specifications:

Schematic:



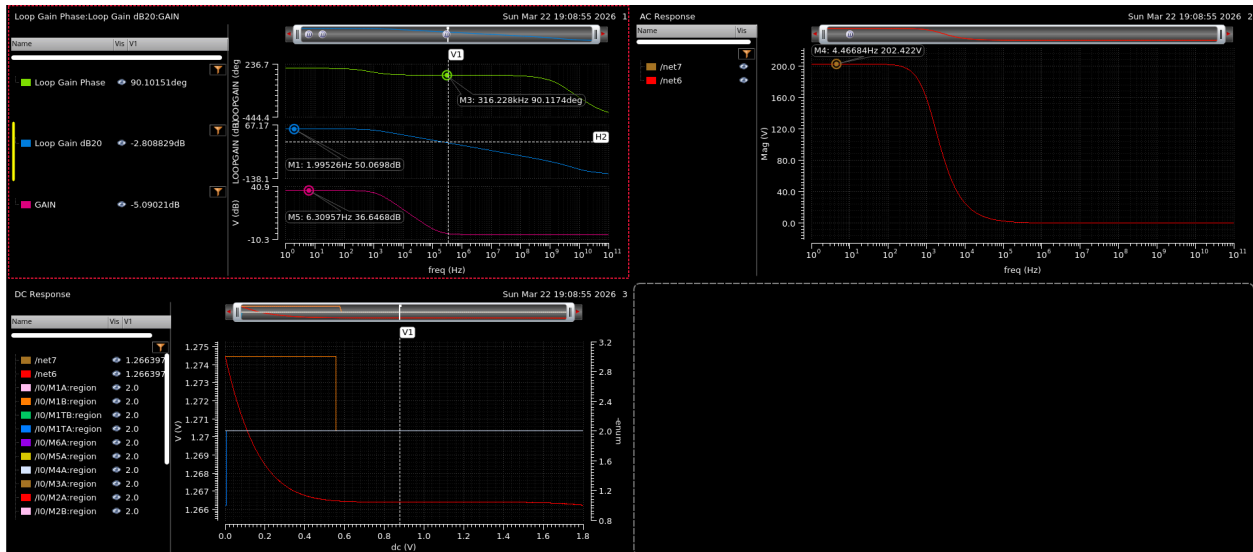
Test-bench setup:



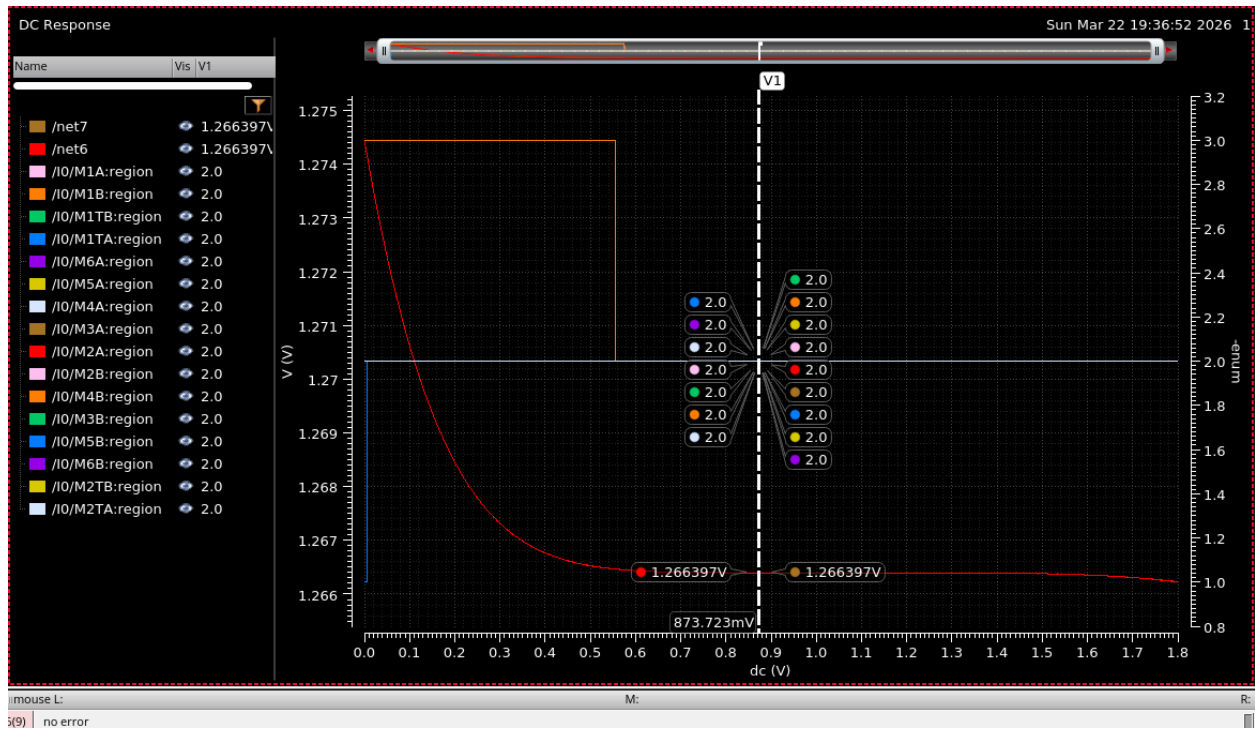


- DC gain > 50 dB

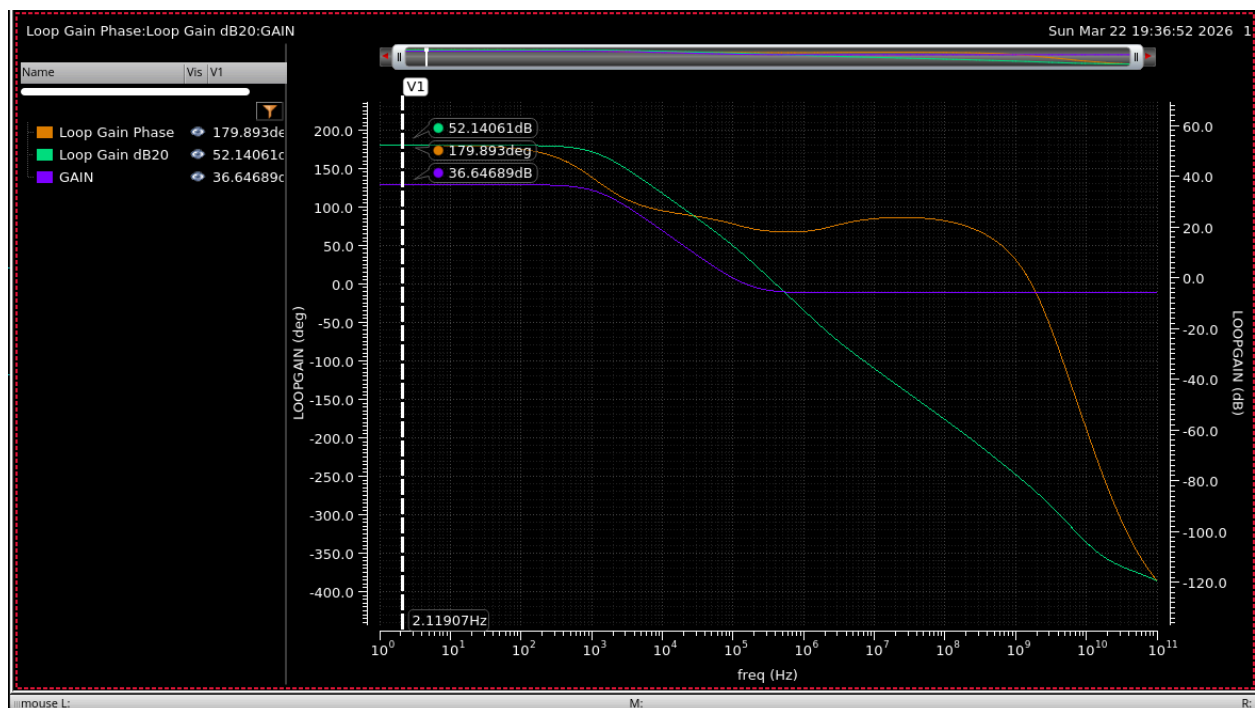
AC Analysis:



Dc analysis after adjusting gain:



Open loop gain and closed loop:

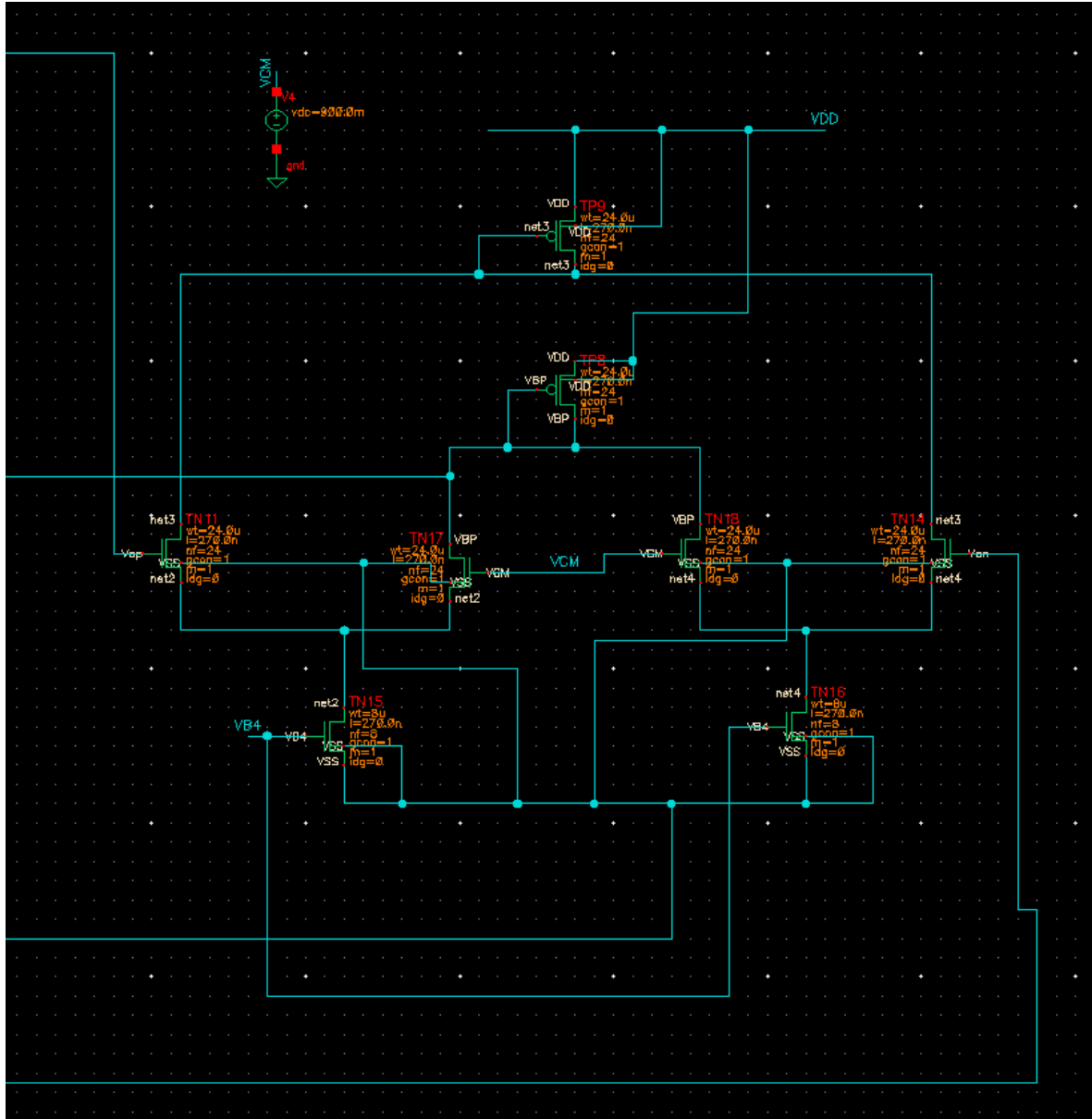


- Total power must be limited to 10 mW; reduce as much as possible the time for slew and linear settling

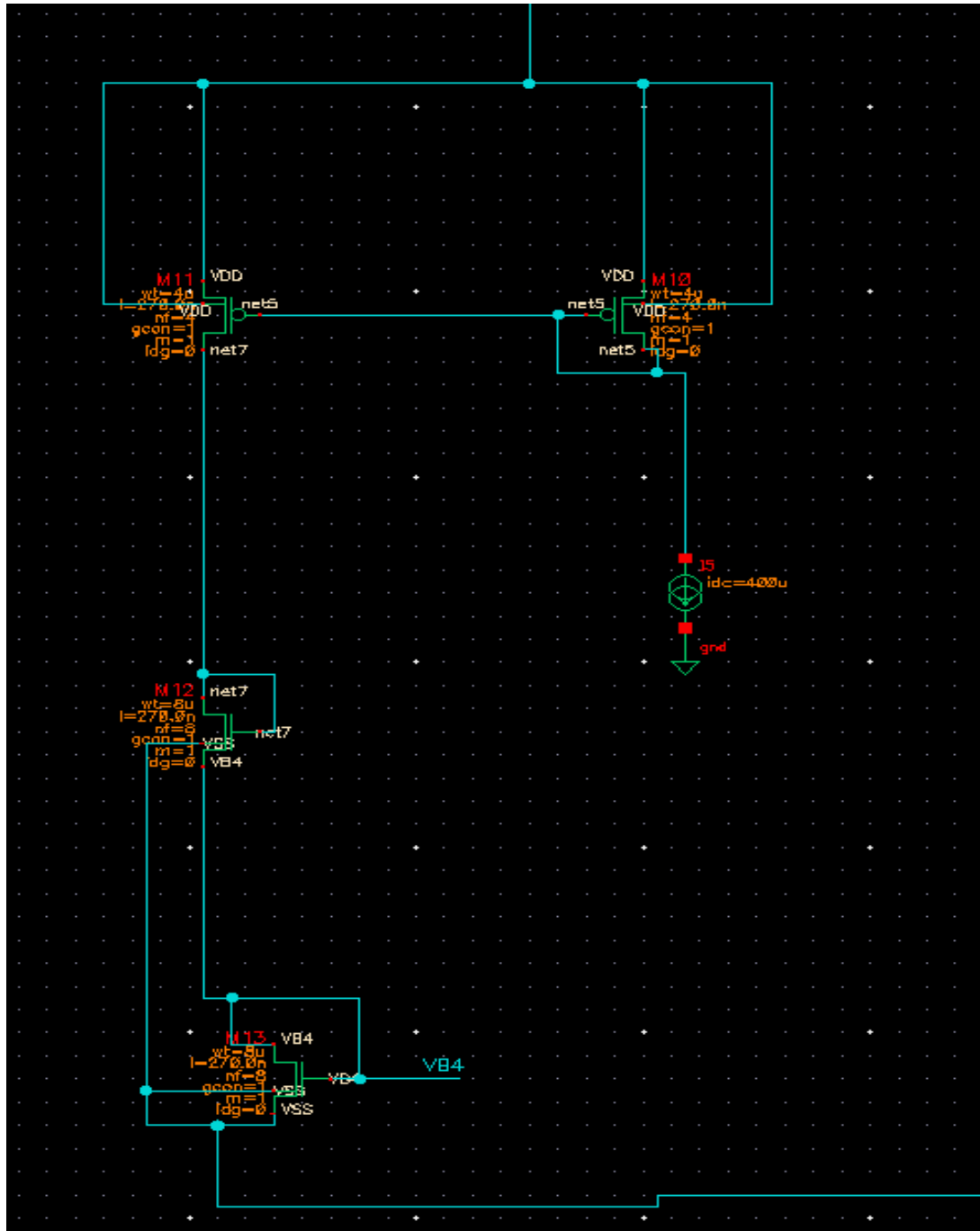
-
- The diagram shows a differential amplifier circuit using two 741 operational amplifiers. Key components and their values are as follows:
- Input Stage:** A differential input signal is applied through a 900mV source (v_2) to the non-inverting inputs of both op-amps. The input resistance is 1.266Ω.
 - Feedback Network:** Each op-amp has a feedback capacitor C_1 (1.266F) and a feedback resistor R_f (1.266Ω). The feedback network also includes a 1.266Ω resistor and a 1.266F capacitor.
 - Output Stage:** The outputs of the two op-amps are connected to a load consisting of a 1.266Ω resistor and a 1.266F capacitor. The output voltage is measured at node 18.
 - Simulation Parameters:** The simulation is set for AC analysis with a frequency range from 1Hz to 1MHz. The input amplitude is 900mV, and the input phase is 0 degrees.
 - Measurements:** The output voltage is measured at node 18, and the differential mode gain is calculated as V_{out}/V_{in} .

- Including cmfb circuit:

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Bias circuit:



i) AC Analysis (Loop Gain):

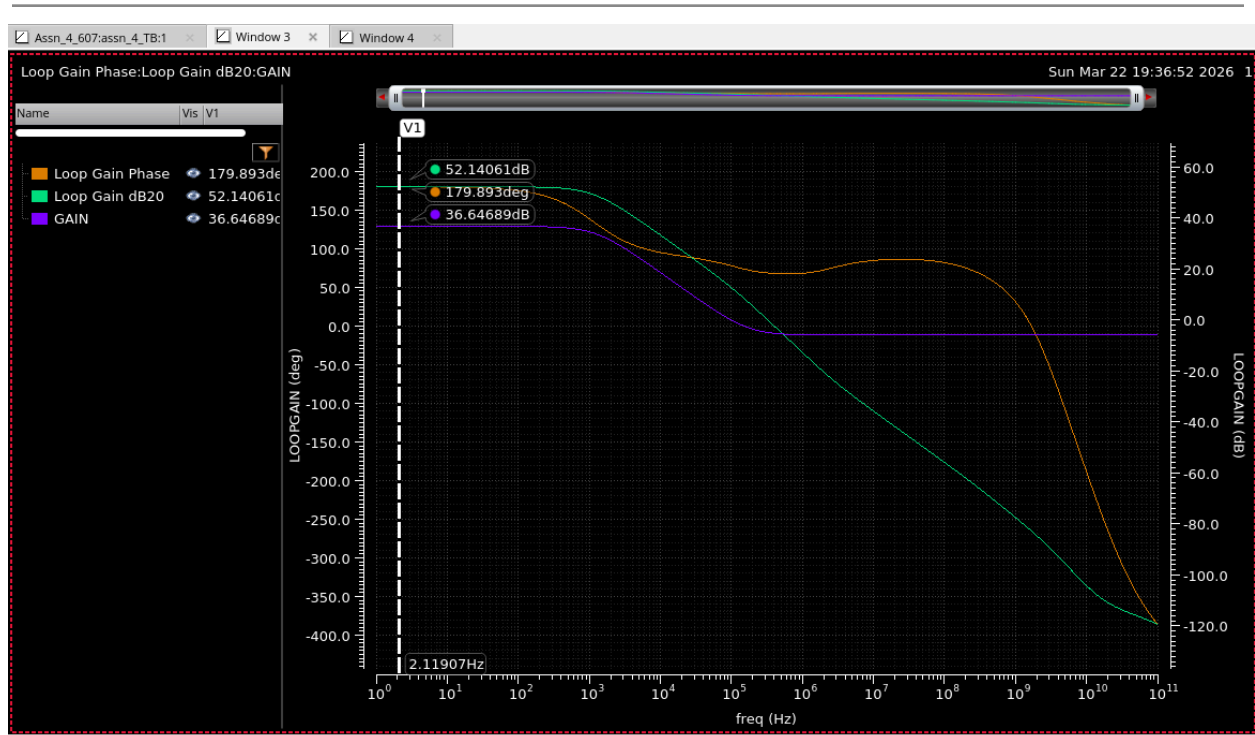
Measure the AC properties of the loop gain.

Do your best to stabilize the architecture.

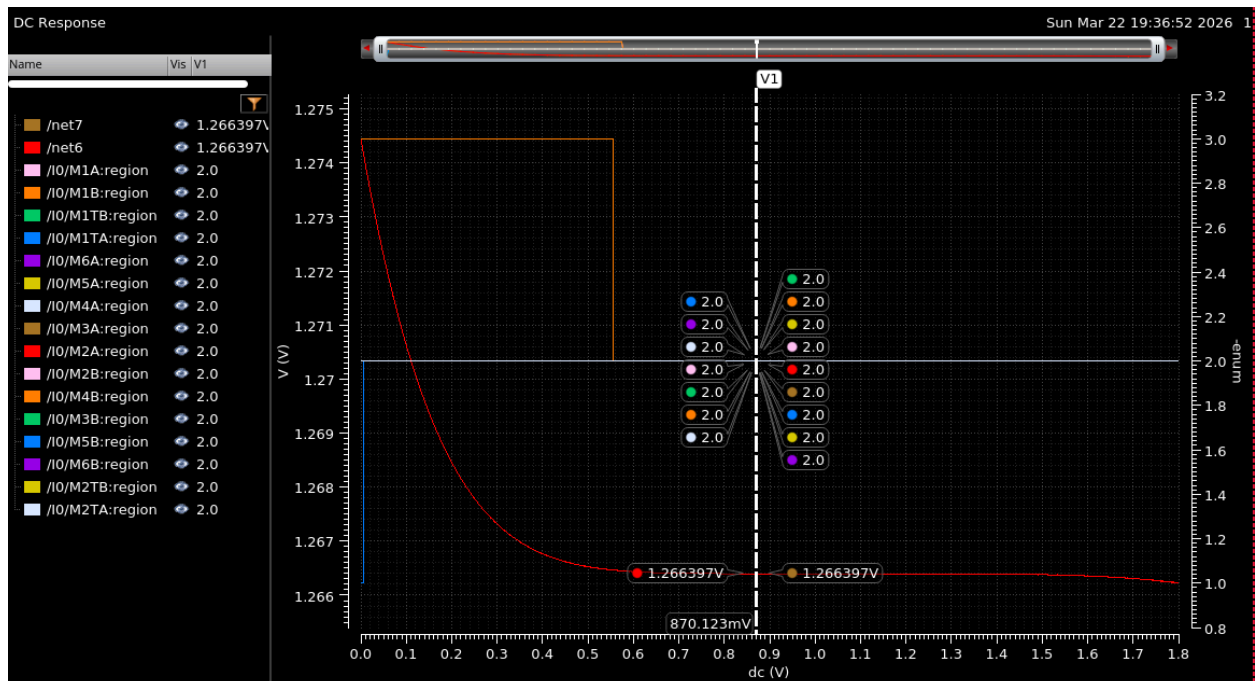
Report:

- Magnitude response
- Phase response

- DC gain
- Phase margin
- Gain margin



DC ANALYSIS AFTER CMFB:



Outputs				
	Name/Signal/Expr	Value	Plot	Save
1	net7		☒	☒
2	net6		☒	☒
3	Loop Gain Phase	wave	☒	☐
4	Loop Gain dB20	wave	☒	☐
5	GAIN	wave	☒	☐
6	net5		☐	☒
7	net4		☐	☒
8	Phase Margin	68.42	☒	☐
9	Gain Margin	82.81	☒	☐
10	GAIN_VALUE	55.45	☒	☐

Parameter	Value
-----	-----
DC Gain	55.45 dB
Phase Margin	68.42°
Gain Margin	82.81 dB

ii) Transient Analysis (Step Response):

Apply a 100 mV step input.

Report:

- Steady-state settling error
- 1% settling time

All settings for transient Analysis:

CDF Parameter	Value	Display
Frequency name for 1/period		off 
Noise file name		off 
Number of noise/freq pairs	0	off 
DC voltage	900m V	off 
AC magnitude		off 
AC phase		off 
XF magnitude		off 
PAC magnitude		off 
PAC phase		off 
Voltage 1	0 V	off 
Voltage 2	100m V	off 
Period		off 
Delay time	1n s	off 
Rise time	10p s	off 
Fall time	10p s	off 
Pulse width	500u s	off 
Temperature coefficient 1		off 
Temperature coefficient 2		off 
Nominal temperature		off 
Type of rising & falling edge		off 

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Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

27

Design Variables

	Name	Value
1	VDD	1.8
2	ID2	70u
3	ID1	70u
4	Vcmdc	900m
5	Vcmac	1
6	Vdmdc	0
7	Vdmac	1
8	Vdmtr	0
9	fdm	0
10	p5vonly	0

Analyses

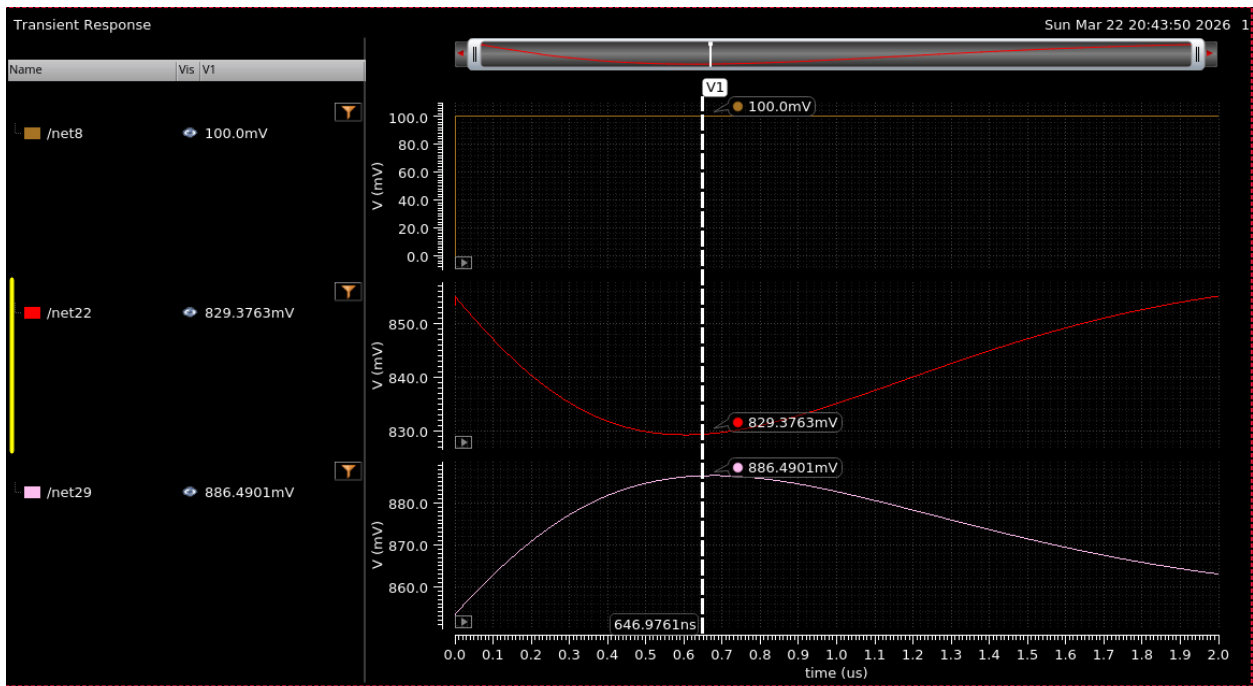
Type	Enable	Arguments
1 dc	☐	t 0 1.8 100u Linear Step Size Start-Stop /V9
2 stb	☐	1 100G 20 /118 Logarithmic Points Per Decade Start-...
3 ac	☐	1 100G 20 Logarithmic Points Per Decade Start-Stop
4 tran	☒	0 2u

Outputs

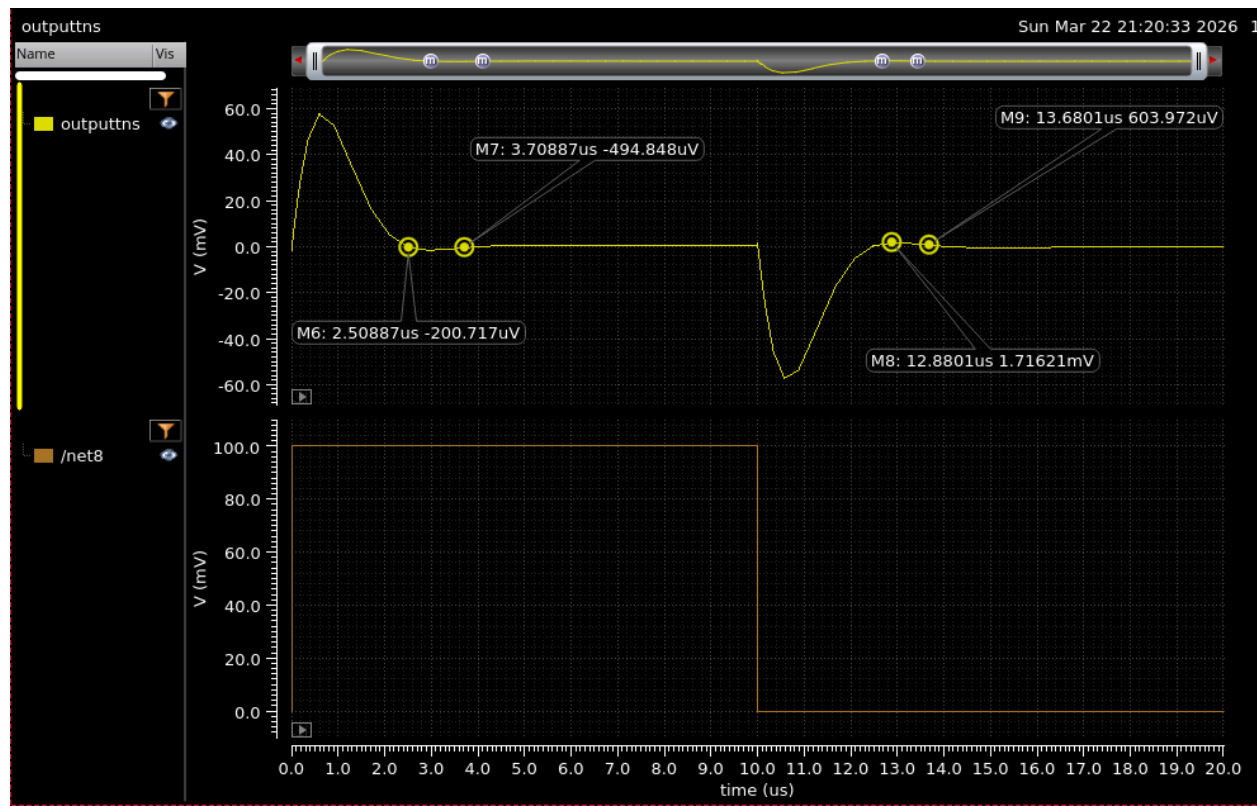
Name/Signal/Expr	Value	Plot	Save	Save Options
6 nets		☐	☒	allv
7 net4		☐	☒	allv
8 Phase Margin		☐	☐	
9 Gain Margin		☐	☐	
10 GAIN_VALUE		☐	☐	
11 net8		☒	☐	allv
12 net22		☒	☐	allv
13 net29		☒	☐	allv

> Results in ..._4_607:assn_4_TB:1/simulation/as

5(6) Load State ... Status: Ready T=27 C Simulator: spectre aps State: Assn_4_607:assn_4_TB:1_active



Transient Analysis:



1% Settling Time Analysis

1. Definition of the 1% Band

For a 100 mV step input:

$$1\% \times 100 \text{ mV} = 1 \text{ mV}$$

Since the final steady-state value is approximately 0 mV, the 1% band is:

$$-1 \text{ mV} \leq V_{\text{out}} \leq +1 \text{ mV}$$

2. Identification of Settling Behavior

From the output transient response:

At marker M8:

Time = 12.8801 μ s, Voltage = 1.71621 mV $\rightarrow$ Outside the 1% band

At marker M9:

Time = 13.6801 μ s, Voltage = 0.603972 mV $\rightarrow$ Inside the 1% band

After M9, the signal remains within the ± 1 mV band.

3. Settling Time Calculation

At M8, the output is slightly above the 1% limit.

At M9, the output enters the 1% band and stays within it thereafter.

Therefore, the 1% settling time is:

$$t_s = 13.68 \mu s$$

4. Summary Table

Marker Time (μs) Voltage Within 1% Band

M6 2.50887 -0.200 mV Yes

M7 3.70873 -0.495 mV Yes

M8 12.8801 1.716 mV No

M9 13.6801 0.604 mV Yes

5. Final Result

The 1% settling time for the given step response is approximately 13.68 μs .

6. Steady-State Error

From the waveform (around 20 μs), the output settles at approximately 0 mV.

Therefore, the steady-state error is approximately zero.

If you want, I can make this even shorter for tight report word limits.

Transistor	Width (W)	Length (L)	W/L
M2TA	16.0 μm (16000 nm)	270.0 nm	59.26
M2TB	16.0 μm (16000 nm)	270.0 nm	59.26

M1A	16.0 μm (16000 nm)	270.0 nm	59.26
M1B	16.0 μm (16000 nm)	270.0 nm	59.26
M6A	1.0 μm (1000 nm)	270.0 nm	3.7
M6B	1.0 μm (1000 nm)	270.0 nm	3.7
M5A	500 nm	270.0 nm	1.85
M5B	500 nm	270.0 nm	1.85
M1TA	55.0 μm (55000 nm)	270.0 nm	203.7
M1TB	55.0 μm (55000 nm)	270.0 nm	203.7
M2A	4.0 μm (4000 nm)	270.0 nm	14.81
M2B	4.0 μm (4000 nm)	270.0 nm	14.81
M3A	220 nm	270.0 nm	0.81
M3B	220 nm	270.0 nm	0.81
M4A	500 nm	270.0 nm	1.85
M4B	500 nm	270.0 nm	1.85

Specifications that are met:

iii) Comparison:

Compare the results of this architecture with those obtained in Assignment 3.

Assignment 3 issues:

No differential output: Single-ended, so no CMRR advantage, more susceptible to supply/common-mode noise.

No CMFB: Cannot control output common-mode for a differential load—OK for single-ended but not for differential design.

DC feedback missing in testbench: In your closed-loop configuration, there's only a capacitor in feedback. This makes DC gain undefined (improper). Steady-state errors are large, and 1% settling times are meaningless—output "returns" to initial value due to missing feedback for DC.

Lower load: 5 pF is significantly larger than 200 fF, so assignment 3 will have a much slower response for the same current.

Assignment 4 advantages:

Fully differential: Superior CMRR, slewing, and symmetric swing.

Includes CMFB: You can accurately set the output common-mode level, which is critical in real-world differential op-amps.

Smaller capacitive loads: (200 fF vs 5pF) enables far quicker response for given current.

Proper DC feedback path (closed loop): Step response settles correctly both in time and amplitude. 1% settling time is clear and measurable.

Better frequency compensation: Phase/gain margins are ample, but not excessive—so speed is maximized for given power.

Transistor sizing: Much larger W/L ratios, supporting higher g_m for differential architecture (and higher bias if needed).

Additional Notes:

- C1, C2, CL connections as specified
- CP must be removed
- Include CMFB for fully differential operation